

Abstract

Static, one-size-fits-all test platforms leave students practicing what they already know while real gaps go unnoticed. The application continuously analyzes responses and dynamically recalibrates question difficulty and topic focus after every attempt, creating focused, personalized revision. It provides topic- and difficulty-wise question banks, instant feedback, performance tracking, and targeted practice recommendations.

Problem Statement & Objectives

- Problem: Static quizzes do not adapt to individual ability or reveal weak areas.
- Objective: Analyze correct/incorrect response patterns in real time.
- Objective: Recalibrate difficulty and topic focus after each attempt.
- Objective: Recommend targeted practice and show progress clearly.

Existing vs Proposed System

Feature	Existing System	Proposed System
Quiz flow	Fixed sequence	Adaptive selection
Personalization	Same for learners	Based on performance
Weak areas	Often missed	Targeted practice
Feedback	Delayed / limited	Instant explanations
Progress	Low visibility	Dashboards & analytics

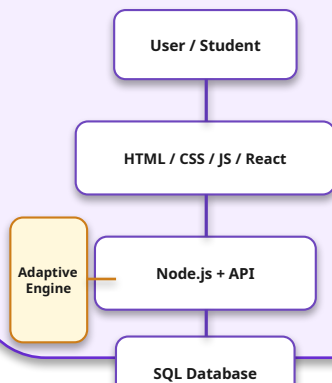
Feasibility Study

- Technical: HTML, CSS, JavaScript, React, Node.js and SQL.
- Economic: Uses common development tools and open-source technologies.
- Operational: Designed for personalized and easy-to-follow practice.
- Schedule: Structured as a semester project with iterative testing.

Tools & Technologies

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js
- Database: SQL
- Version Control: GitHub
- Development: VS Code

System Architecture



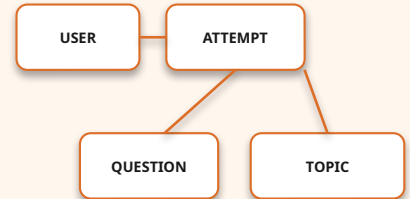
Methodology

1. Requirement Analysis — define adaptive quiz logic and roles.
2. UI/UX & SQL Design — plan screens and database schema.
3. Frontend Development — build responsive interfaces.
4. Backend & API Development — implement server-side logic.
5. Adaptive Engine & Testing — integrate and test adaptation.
6. Deployment & Monitoring — observe performance post-launch.

Results & Testing

- Adaptive selection recalculates question difficulty/topic focus.
- Correct/incorrect patterns support targeted recommendations.
- Instant explanations and dashboards provide progress visibility.

ER Diagram & Database Design



Conceptual entities derived from the adaptive quiz workflow.

Key Features

- Adaptive algorithm recalibrates question selection.
- Topic-wise and difficulty-wise question banks.
- Instant feedback with detailed explanations.
- Performance analytics and progress tracking.
- Targeted practice for weaker concepts.

Implementation Modules

LOGIN	Secure user access
QUIZ	Adaptive question flow
ANALYTICS	Performance dashboard
FEEDBACK	Explanations & guidance

Conclusion

- Focused learning: practice is directed toward weak areas.
- Real-time adaptation: difficulty and topics change with performance.
- Measurable growth: dashboards make improvement visible.
- Full-stack implementation aligns with course objectives.

Future Scope

- Expand subject/question-bank coverage.
- Enhance analytics and personalized recommendations.

